A Convex-Cone Reachability Oracle for Cell-State Engineering: Falsifiable Feasibility Verdicts and Constructive Certificates from Measured Genome-Scale Perturbation Screens

Cell-State Reachability Project [1]

Abstract

Deciding whether a desired change in cell state is achievable with a given class of perturbation—and certifying when it is not—is central to cell-state engineering and drug-target selection, yet existing methods only predict responses or rank regulators, and none returns a feasibility verdict grounded in measured effects. We introduce a convex-cone reachability oracle that treats a genome-scale CRISPRi Perturb-seq screen as a dictionary of measured effect vectors and, because knockdowns combine additively and non-negatively, decides by non-negative least squares whether a target cell-state shift lies in their conic hull. A reachable target yields a minimal ranked knockdown recipe, while an unreachable one yields a Farkas/KKT separating-hyperplane certificate that names the genes which must instead be *activated*—a falsifiable gain-of-function hypothesis rather than a null result. In primary human CD4$^+$ T cells the $\mathrm{T_H2}{\to}\mathrm{T_H1}$ polarization shift is only *partly* reachable by knockdown (held-out cosine 0.448 against a shuffled-target null, $z \approx 24$), and across a 12-cell atlas knockdown is never the majority modality (mean loss-of-function fraction 0.34), so an honest method must report that fraction rather than a single reachable/unreachable bit. The identical operator transfers without retuning to a K562 CRISPRa screen (held-out CEBPA cosine 0.878), whose measured double perturbations show that the additivity assumption is a bounded approximation (median cosine 0.71) rather than an identity. Against a survey of 92 methods (2011–2026) measured-effect grounding and a certificate-carrying verdict are held by disjoint method sets with empty intersection, so we position the oracle as an experiment-triage instrument—deciding where a screen should spend its arms and where it should stop—whose every nomination is a falsifiable wet-lab hypothesis, not a validated target.

1. Introduction

Much of experimental biology is a search for interventions that move a cell from one state to another: differentiating a stem cell, repolarizing a T cell, reverting a diseased phenotype, or—in drug discovery—knocking down a target to reverse a pathogenic program. Before committing reagents to such a program, the decisive question is not *what will this perturbation do* but *can the desired change be achieved at all with the perturbations available, and if not, what is missing*. That question is rarely asked computationally, and its economic weight is large: only about one in ten clinical programs succeeds, most failures are for lack of efficacy rather than safety, and the interventions with demonstrated power to shift those odds—human genetic support and directly measured target effects—both argue for grounding target choice in real perturbation data rather than inferred models (Minikel et al., 2024; Nelson et al., 2015; Harrison, 2016; Begley & Ellis, 2012; Prinz et al., 2011).

The computational tools available to answer it fall into two camps that, as we document in Section 4, do not overlap. Perturbation-response models—from gene-regulatory-network simulators to single-cell foundation models—are increasingly trained on real screens but are forward predictors: given a perturbation they return a predicted response, and a predictor structurally always returns *some* answer, so it cannot certify that a goal is out of reach. Network-control and Boolean-attractor methods do reason about reachability and can carry controllability guarantees, but those guarantees are properties of a hand-built or inferred model, only as trustworthy as the network assumed. Neither camp answers the feasibility question from measured effects. This gap matters all the more because a wave of 2025–2026 benchmarks finds that current perturbation predictors, foundation models included, frequently fail to beat mean or linear baselines on genuinely unseen perturbations (Ahlmann-Eltze

[1] Built with Claude — Life Sciences (Research Track). Correspondence to: Cell-State Reachability Project <github.com/MasalaKimchi/cell-state-reachability>.

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et al., 2025; Wong et al., 2025; Kernfeld et al., 2025; Atti & Subramaniam, 2025): the field's own evidence argues for an output weaker and more defensible than predicting an unseen response.

The convex-cone move. We take the opposite starting point from model building: we do not infer a network at all. A genome-scale CRISPRi Perturb-seq screen (Zhu et al., 2025) directly measures, for each of tens of thousands of single-gene knockdowns, the effect vector it produces in expression space. We treat these measured vectors as a fixed dictionary and ask a geometric question: does the desired cell-state shift lie in the set of shifts the dictionary can produce? Two facts about knockdowns make this question exactly a convex-geometry problem. A knockdown can only remove function, fixing the sign of each atom's contribution; and applying several knockdowns together produces, to first order, an additive combination with non-negative loadings. The set of achievable shifts is therefore the *non-negative conic hull* of the measured effect vectors, and asking whether a target is reachable is asking whether it lies inside that cone—a question non-negative least squares (NNLS) answers exactly (Lawson & Hanson, 1974). This reframing turns an open-ended design search into a decision with a proof on both sides. When the target is inside the cone, the NNLS weights *are* a minimal ranked knockdown recipe. When it is outside, convex duality supplies a separating hyperplane—a Farkas/KKT certificate (Farkas, 1902; Kuhn & Tucker, 1951)—whose positive coordinates name the genes the target requires but that no combination of knockdowns can supply. An infeasibility result is thus not a dead end but a constructive, falsifiable hypothesis: it says the move requires *activation*, and it names the genes to activate, redirecting the experiment from a CRISPRi arm to a CRISPRa arm.

Scope. The oracle is a decision instrument, not an oracle of biological truth: it certifies feasibility *relative to the measured dictionary* under an explicit additivity model, and its verdicts are hypotheses for the bench, not validated targets. The regulators it surfaces in T cells and in the CEBPA program are already reported by the source screens (Zhu et al., 2025; Norman et al., 2019) and enter here only as positive controls; the novelty we claim is the measured-effect, certificate-carrying reachability *decision* itself, not the biology it recovers.

Contributions. This paper makes four contributions.

1. **A feasibility oracle for cell-state engineering.** We formalize reachability of a target cell-state shift as membership in the convex cone of measured perturbation effect vectors, and solve it with NNLS so that a verdict, a minimal ranked recipe, and—on infeasibility—a Farkas/KKT activation certificate all fall out of one convex program (Section 2).
2. **A validated headline result.** On primary human CD4+ T cells the oracle finds $\mathrm{T_H}2 \to \mathrm{T_H}1$ partly reachable by knockdown—far above a shuffled-target null but well below full reachability—with a signed loss-/gain-of-function decomposition and canonical regulators recovered as positive controls (Section 3.1).
3. **Generality across transitions and datasets.** A 12-cell atlas shows knockdown is never the majority modality across four transitions and three activation states, and the identical operator transfers without retuning to a K562 CRISPRa combinatorial screen, where we also quantify the additivity approximation directly on measured doubles (Sections 3.2–3.4).
4. **An exact positioning against the literature.** Over 92 methods spanning 2011–2026, we show that measured-effect grounding and feasibility-with-certificate are held by disjoint method sets whose intersection is empty, and translate the resulting verdicts into a GO/REDIRECT/STOP triage over 102 real target nominations (Sections 3.5–3.6).

2. Methods

2.1. Data and effect vectors

The primary dictionary is the genome-scale CRISPRi Perturb-seq screen of Zhu et al. (2025) in primary human CD4+ T cells (Marson and Pritchard laboratories, CZI Virtual Cells Platform), comprising differential-expression statistics for 33,983 single-gene knockdowns across 10,282 measured genes from four donors. For each knockdown j we take the published per-gene effect vector $\mathbf{E}_{:,j} \in \mathbb{R}^G$, the signed transcriptional shift the knockdown induces relative to non-targeting controls. Stacking these columns gives the effect matrix $\mathbf{E} \in \mathbb{R}^{G \times P}$. A *target* is a desired cell-state shift $\mathbf{d} \in \mathbb{R}^G$, computed as the difference between the mean expression profile of the source state and that of the destination state (e.g. $\mathrm{T_H}2$ minus $\mathrm{T_H}1$). All effect and target vectors are restricted to the genes measured in both the screen and the target contrast, and comparisons are reported as cosine similarity so that magnitude scaling does not affect the verdict. The identical pipeline is applied without modification to the K562 combinatorial CRISPR screen of Norman et al. (2019) (Section 3.4); no dataset-specific parameters are tuned.

2.2. Reachability as membership in a convex cone

Two properties of loss-of-function perturbations determine the geometry. First, a CRISPRi knockdown removes rather than adds function, so each atom $\mathbf{E}_{:,j}$ enters a combination with a fixed orientation. Second, applying a set of knockdowns together produces, to first order, the sum of their individual effects; there is no natural notion of a "negative amount" of a knockdown. The set of cell-state shifts reachable by a non-negative combination of knockdowns is therefore the *non-negative conic hull* of the measured effect vectors,

$$\mathcal{C} \;=\; \left\{ \mathbf{E}\mathbf{w} \;:\; \mathbf{w} \in \mathbb{R}^{P}_{\geq 0} \right\} \;\subseteq\; \mathbb{R}^{G}, \tag{1}$$

a closed convex cone. Deciding whether a target $\mathbf{d}$ is reachable is deciding whether $\mathbf{d} \in \mathcal{C}$ (up to a non-negative scale, since only direction is claimed). We quantify proximity by projecting $\mathbf{d}$ onto $\mathcal{C}$: the reachable component is the solution of the non-negative least-squares (NNLS) program

$$\mathbf{w}^\star \;=\; \arg\min_{\mathbf{w} \geq 0} \left\| \mathbf{E}\mathbf{w} - \mathbf{d} \right\|_2^2, \qquad \hat{\mathbf{d}} \;=\; \mathbf{E}\mathbf{w}^\star, \tag{2}$$

solved with the Lawson–Hanson active-set algorithm (Lawson & Hanson, 1974). The *reachable cosine* $\cos(\hat{\mathbf{d}}, \mathbf{d})$ measures how much of the target's direction the cone can express, and the residual $\mathbf{d} - \hat{\mathbf{d}}$ is the part it cannot. A target is declared reachable when its reachable cosine is both high in absolute terms and far above the null of Section 2.5; the same solve returns the support of $\mathbf{w}^\star$, whose largest entries constitute a *minimal ranked knockdown recipe*. Ordering genes by a greedy forward selection that adds at each step the knockdown giving the largest marginal gain in reachable cosine yields the reachability spectrum reported throughout.

2.3. Infeasibility certificates via convex duality

The value of a feasibility oracle is that "no" is provable, not merely observed. When $\mathbf{d} \notin \mathcal{C}$, the separating-hyperplane theorem for convex cones (a form of Farkas' lemma (Farkas, 1902)) guarantees a vector $\boldsymbol{y}$ with $\mathbf{E}^\top \boldsymbol{y} \leq 0$ and $\mathbf{d}^\top \boldsymbol{y} > 0$: a hyperplane with the whole cone on one side and the target strictly on the other. This $\boldsymbol{y}$ is exactly the negative gradient of the NNLS objective at the optimum, so it is produced by the same solve rather than a separate computation. The Karush–Kuhn–Tucker (KKT) stationarity conditions of (2) provide the numerical proof: at the optimum, active (positive-weight) atoms have zero reduced gradient and inactive atoms have non-positive reduced gradient. We report the maximum stationarity violation as the certificate residual; across all reported solves it is at machine precision ($\approx 1.1 \times 10^{-11}$), confirming true optimality rather than early stopping. The biological reading of the certificate is constructive: the positive coordinates of the residual direction—the part of the target orthogonal to everything the cone can supply—name genes the target requires to be *raised* that no knockdown combination can deliver. These become falsifiable CRISPRa (activation) hypotheses, turning an infeasibility result into a redirected experiment rather than a null one.

2.4. Signed loss-/gain-of-function decomposition

To separate "cannot be reached because it needs activation" from "cannot be reached for other reasons," we decompose the target into three orthogonal budgets as fractions of its squared norm: the component captured by the non-negative knockdown cone (loss-of-function, LOF); the component lying along directions that require increased expression (gain-of-function, GOF), obtained by projecting the residual onto the positive-expression orthant; and an irreducible remainder aligned with neither ("neither"). We report this in two conventions. The *signed* decomposition partitions the target by the sign of each coordinate's required change—an orthogonal split along mutually polar cones in the sense of Moreau (1962); the *one-sided* decomposition instead attributes the residual after the non-negative projection. For $\mathrm{T_H}2 \to \mathrm{T_H}1$ these give LOF/GOF/neither of $39\%/25\%/35\%$ (signed) and $39\%/31\%/30\%$ (one-sided). We report both because they answer subtly different questions and neither is privileged; the qualitative conclusion—that loss-of-function alone accounts for well under half of the shift—is common to both.

2.5. Held-out validation and null models

Two null models guard against two distinct failure modes, each addressing a different question.

Held-out-gene generalization. An in-sample reachable cosine can be inflated by a solver freely reweighting thousands of atoms to fit a fixed target; it measures expressivity of the dictionary, not predictive validity. We therefore report a *held-out-gene* cosine: we fit the NNLS weights using only a random subset of measured genes (rows of $\mathbf{E}$ and $\mathbf{d}$) and evaluate the reachable cosine on the held-out genes, so the recipe must generalize to expression coordinates it was not fit on. For $\mathrm{T_H}2 \to \mathrm{T_H}1$ the held-out cosine is 0.448, against an in-sample value of 0.627; the gap is the cost of generalization, and we quote the held-out number as the headline throughout.

Shuffled-target null. To calibrate what reachable cosine arises by chance from a high-dimensional cone, we recompute the held-out reachability against 60 random targets generated by shuffling the gene labels of $\mathbf{d}$, preserving its marginal magnitude distribution while destroying its biological structure. For $\mathrm{T_H}2 \to \mathrm{T_H}1$ the null has mean -0.005, standard deviation 0.019, and maximum 0.029; the observed held-out cosine sits at $z \approx 24$ above it. This null answers "is the target special," distinct from held-out-gene generalization, which answers "does the recipe transfer to unseen coordinates."

Analytic anisotropy null. Because a shuffled-target null over an atlas of targets is expensive, we derive a closed-form approximation to the expected null cosine as a function of two geometric quantities—the alignment of the target with the dominant direction of the cone and the effective anisotropy of the atom cloud (Supplementary Fig. S1 legend). Validated against the shuffle null on both synthetic and real CD4+ data, it reproduces the empirical null to Pearson $r = 0.995$ (in-sample) and 0.998 (held-out), with maximum absolute error 0.047 and 0.07 respectively, replacing on the order of a thousand target shuffles with ten to twenty NNLS solves. This makes the atlas-wide reliability analysis of Section 3.2 tractable while keeping the empirical shuffle null as the reference for the headline result.

Positive controls. We verify that the geometry recovers known biology using canonical T-helper regulators as controls, *not* as discoveries. On the toward-$\mathrm{T_H}1$ track, GATA3—the master $\mathrm{T_H}2$ transcription factor, which must be removed—is reached at rank $155/6871$ (97.7th percentile, cosine $+0.101$) on the loss-of-function side, while TBX21—the master $\mathrm{T_H}1$ factor, which must be raised—is correctly anti-aligned under knockdown at rank $6775/6871$ (1.4th percentile). Because these regulators are already reported by the source screen (Zhu et al., 2025), their recovery validates the operator without being claimed as a novel finding.

2.6. Robustness and reinforcement analyses

Five analyses probe whether the headline verdict is an artifact of a specific modelling or evaluation choice. All reuse `reachability.py` without modification.

Non-negativity ablation. To test whether the biological non-negativity constraint earns its place, we compare the held-out cosine of the constrained NNLS solve against an unconstrained least-squares solve (which permits negative weights, i.e. treating a knockdown as if it could be run in reverse) and against the single best-aligned generator, across all 12 atlas cells. We report the cosine cost of the constraint, the number of biologically unrealizable negative weights the unconstrained solution requires, and the fraction of the unconstrained cosine the cone recovers.

Reachable-cosine ceiling. Because a knockdown-only cone cannot express gain-of-function directions, the achievable held-out cosine has a modality-imposed ceiling. We compute this ceiling as $\sqrt{f_{\mathrm{LOF}}}$, where f_{LOF} is the loss-of-function fraction of the signed decomposition, and express each headline cosine as a fraction of its own ceiling so that a "modest" absolute number is reported against what is attainable in principle rather than against 1.

DEG-magnitude weighting. Uniform-gene metrics can dilute a real signal by averaging over thousands of unchanged genes, a critique raised for perturbation-response evaluation by Mejia et al. (2025). We recompute every reachable cosine and held-out z under a weighting $w_g = |d_g|$ that concentrates the metric on the genes the target actually moves, and we recalibrate the dynamic-range fraction against a shuffled-gene floor and an interpolated-duplicate ceiling under the same weighting. The unweighted metric remains the default and reproduces the published numbers exactly; the weighting is a stress test, not a re-definition.

Cross-cell-type transfer. To ask whether the operator generalizes beyond a single cell type, we apply it to the genome-scale essential-gene Perturb-seq screens of Replogle et al. (2022) in K562 and RPE1 cells (843 shared single-gene perturbations over $2{,}832$ shared genes). We evaluate transfer at three levels: whether a single perturbation's effect *direction* agrees across cell types (matched cosine against a shuffled-gene null); whether the reachability *verdict* agrees (graded by the residual norm within- versus cross-basis, since a binary verdict is uninformative when an 843-atom cone is highly expressive); and whether the minimal *recipe* agrees (Jaccard overlap of selected atoms against a random-selection null).

Certificate stability and external cross-reference. We test the activation certificate two ways. For reproducibility, we refit the certificate on a random half of the gene axis and measure the rank agreement of the top genes against the full-data ranking and across independent halves (Spearman ρ over 12 random splits, against a label-shuffle null). For biological interpretation, we retrospectively cross-reference the top certificate genes against the literature direction-of-effect and against Open Targets genetic-association scores for the relevant $\mathrm{T_H}1$-driven autoimmune diseases. This cross-reference is an exploratory annotation, not a pre-registered test, and we report it as such.

2.7. Software and reproducibility

The oracle is implemented in a single module (`reachability.py`) built on standard NNLS routines; all reported quantities derive from the published effect matrices of Zhu et al. (2025) and Norman et al. (2019) with no additional model fitting. Every numeric value in this manuscript is drawn from a locked facts sheet tied to the analysis outputs, and each figure is generated directly from those outputs; figures were checked for exact correspondence to the facts sheet before inclusion.

3. Results

Figure 1 summarizes the framework: a genome-scale screen yields measured effect vectors (1A); reachability is membership in their convex cone, with the residual naming what knockdown cannot supply (1B); a reachable target returns a verdict, a signed decomposition, and an activation certificate (1C); and the same operator transfers to an independent screen (1D).

3.1. The $\mathrm{T_H}2 \rightarrow \mathrm{T_H}1$ shift is partly reachable by knockdown

Our headline target is the $\mathrm{T_H}2 \rightarrow \mathrm{T_H}1$ polarization shift in resting cells, a canonical and therapeutically relevant repolarization. The greedy reachability spectrum (Fig. 2a) rises rapidly and plateaus: the reachable cosine reaches a knee at $k = 7$ knockdowns (cosine 0.40) and saturates near 0.45, sitting far above the shuffled-target null at every recipe size. The held-out-gene cosine is 0.448 against a null of mean -0.005 and standard deviation 0.019 ($z \approx 24$), so the target is unambiguously reachable in the statistical sense—the recipe generalizes to expression coordinates it was not fit on—yet the absolute value is well below 1. This is the central result: a real biological repolarization is *partly*, not fully, reachable by loss-of-function alone.

The signed decomposition (Fig. 2b) explains why. Only 39% of the target shift lies in the knockdown cone; 25% requires gain-of-function, and the remaining 35% aligns with neither. The activation certificate (Fig. 2c) makes the gain-of-function budget concrete, naming seven genes—LYAR, IKZF3, CRTAM, CBLB, GBP5, LAG3, and IRF8—that the $\mathrm{T_H}1$ target requires to be raised but that no non-negative combination of knockdowns can deliver. These are not a null result but a set of falsifiable CRISPRa hypotheses: they specify exactly which activations a follow-up experiment should test.

The geometry recovers known biology as a positive control (Fig. 2d). GATA3, the master $\mathrm{T_H}2$ transcription factor that must be removed to become $\mathrm{T_H}1$, is reached on the loss-of-function side at rank $155/6871$ (97.7th percentile). TBX21, the master $\mathrm{T_H}1$ factor that must instead be *raised*, is correctly anti-aligned under knockdown at rank $6775/6871$ (1.4th percentile)—the oracle refuses to knock down the gene the target needs induced. Both regulators are already reported by the source screen (Zhu et al., 2025); their recovery validates the operator and is not claimed as a discovery. The first genes entered by the greedy recipe (LAT2, APPBP2, RARA, ICOS) are likewise consistent with T-cell signaling and differentiation biology rather than novel nominations.

3.2. Knockdown is never the majority modality across a 12-cell atlas

To test whether the partial-reachability finding is idiosyncratic to one transition, we ran the oracle across an atlas of 12 target cell-states: four transitions (toward $\mathrm{T_H}1$, toward $\mathrm{T_H}2$, toward younger, toward older) under three activation conditions (resting, stimulated $8\,\mathrm{h}$, stimulated $48\,\mathrm{h}$). Every one of the 12 targets is statistically reachable, with held-out z ranging from 4.7 to 45.0; none is a false positive of the cone's dimensionality. But in no case does loss-of-function account for the majority of the shift (Fig. 3a): the mean loss-of-function fraction is 0.34, and the knockdown bar never crosses the 50% line. Resting cells are the most reachable along all four axes—consistent with activation adding gain-of-function directions that knockdown cannot span—and the verdicts range from "partially reachable" (four resting transitions) to "weakly reachable" (the stimulated conditions). The general conclusion is robust: across transitions, cell states, and activation conditions, engineering by knockdown alone reaches a real but minority fraction of the desired change, and a method must report that fraction rather than a single reachable/not-reachable bit.

The atlas also exposes a calibration hazard (Fig. 3b). The in-sample reachable cosine (mean 0.580) systematically overstates the held-out-gene cosine (mean 0.355), by a mean gap of 0.225 (range 0.18–0.27) across all 12 transitions; an 85% without-replacement gene-panel bootstrap gives tight confidence intervals (e.g. toward $\mathrm{T_H}1$ resting: point 0.627, bootstrap mean 0.639, CI $[0.633, 0.649]$), confirming that the gap is a real optimism bias and not estimation noise. We therefore assign a confidence tier to each transition from its held-out z (2 high, 6 medium, 4 low) and quote held-out numbers everywhere. Finally, the closed-form anisotropy null lands almost exactly on the empirical shuffle null (Fig. 3c; Pearson $r = 0.995$ in-sample, 0.998 held-out), which is what makes the atlas-wide analysis affordable.

3.3. The verdict is robust to its modelling and scoring choices

The verdict does not rest on any single modelling or scoring choice behind it. The biological non-negativity constraint is nearly free yet does real work: relative to an unconstrained least-squares fit that may run knockdowns in reverse it costs a mean of only 0.018 in held-out cosine, but the unconstrained fit buys that margin with $\approx 3{,}537$ biologically unrealizable negative weights, so the cone recovers a mean 95% (range 86–103%) of the unconstrained cosine, beats the best single generator in all 12 cells, and is the sole source of the Farkas/KKT certificate. The modest absolute cosine is in turn close to the ceiling the modality allows: because a knockdown-only cone cannot express gain-of-function directions, the attainable held-out cosine is bounded by $\sqrt{f_{\mathrm{LOF}}}$, and the headline 0.448 is 71% of that bound (atlas mean 61%, range 49–71%), leaving the residual 22–26% gain-of-function-locked—most of what loss-of-function can supply, not a tuning shortfall. This bound is not an independent check: $\sqrt{f_{\mathrm{LOF}}}$ coincides with the in-sample reachable cosine (0.627 here, to 10^{-4}), so "71% of the ceiling" is the same statement as the held-out cosine recovering 71% of the in-sample fit. The recommended recipes stay well inside the additivity regime that justifies them, too: every knee

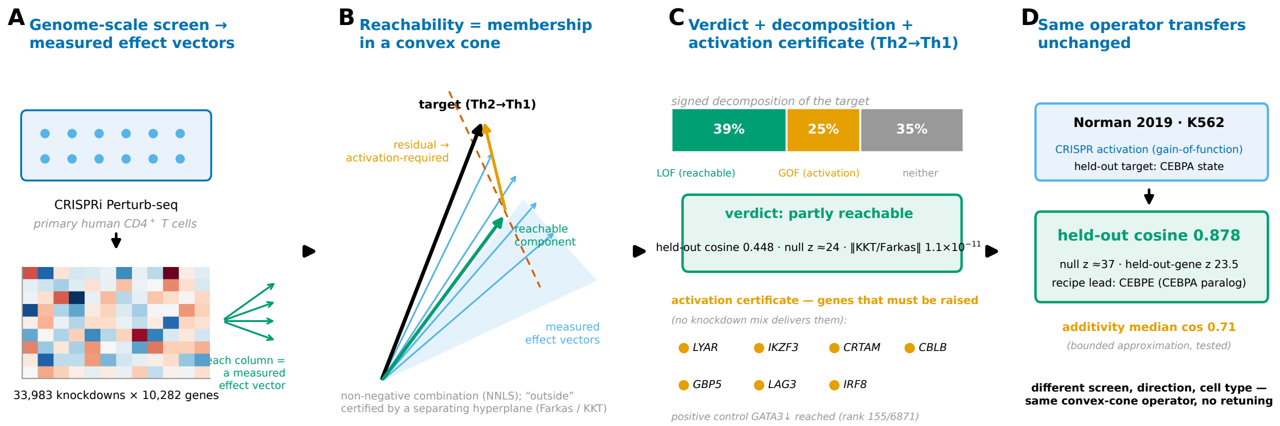

Figure 1. **The convex-cone reachability oracle. (A)** A genome-scale CRISPRi Perturb-seq screen in primary human CD4$^+$ T cells yields, for each of 33,983 knockdowns, a measured effect vector over 10,282 genes. **(B)** Reachability is membership in the non-negative conic hull of these vectors; the reachable component is the NNLS projection of the target, and the residual is the activation-required direction, with "outside" certified by a separating hyperplane (Farkas/KKT). **(C)** For $T_H2 \to T_H1$ the oracle returns a *partly reachable* verdict (held-out cosine 0.448, $\|$KKT/Farkas$\| \approx 1.1 \times 10^{-11}$), a signed decomposition (39% LOF / 25% GOF / 35% neither), and an activation certificate naming genes that must be raised. **(D)** The identical operator, with no retuning, transfers to the Norman K562 CRISPRa screen (held-out CEBPA cosine 0.878).

recipe is additive-safe across all 12 cells (per-recipe reliability 0.92–0.96), and the combined-magnitude saturation cap binds only near $k \approx 28$ knockdowns against a knee of $k \approx 4$–5, a 5.6× margin on the headline recipe.

Two sharper challenges aimed at the evaluation itself leave the conclusion stronger rather than weaker. The signal-dilution critique of Mejia et al. (2025)—that a uniform-gene metric averages a real effect away over unchanged genes—would deflate the reachable cosine if it applied; reweighting the metric by each gene's target magnitude ($w_g = |d_g|$) instead *raises* the Tier-2 $T_H2 \to T_H1$ cosine from 0.627 to 0.803 at rest (and every stimulated condition and held-out Norman double likewise), with the held-out z staying far above the significance floor (28.3 $\to$ 14.1 at rest; Fig. S5). The calibrated placement between a shuffled floor and an interpolated-duplicate ceiling barely moves (dynamic-range fraction 0.447 $\to$ 0.499) because the floor rises in step (0.346 $\to$ 0.619), so the gain is genuine rather than a scaling artifact (Fig. S6), and the unweighted metric—still the default—reproduces the published numbers exactly. The certificate, likewise, is reproducible where it should be and guarded where it cannot yet be otherwise: under a random gene-axis split its top genes hold rank (cross-half Spearman $\rho = 0.65$, half-versus-full 0.84, $z \approx 35$ against a ≈ 0 shuffle null, full top-30 retained) while the tail drifts (Figs. S8–S9). A retrospective cross-reference of the top 20 tempers their reading: the highest-demand genes are state-markers (LYAR, CRTAM, GBP5) and *negative* regulators (IKZF3/Aiolos, CBLB, LAG3), whose *deletion* rather than activation boosts T-cell function, while the canonical T_H1 master transcription factors sit deep in the list (IFNG #35, TBX21 #144, STAT4 #364, STAT1 #473); of the 20, four are literature-supported, seven plausible, nine unknown, and 8/20 carry a strong (≥ 0.5) Open Targets genetic link to a T_H1-driven autoimmune disease. The certificate is thus a reproducible ranking of unmet upward demand rather than a validated activation list, and for the negative regulators a useful intervention may run opposite to naive activation—a qualification we carry into the discussion.

Applying the operator to an independent pair of cell types bounds what transfers, which we develop next.

3.4. The operator transfers unchanged to a K562 CRISPRa screen

To show the operator is not bespoke to one screen, we applied the identical pipeline—same code, no retuned parameters—to the K562 combinatorial CRISPR activation screen of Norman et al. (2019) (GSE133344), a different cell type, a different perturbation direction (activation rather than interference), and a different experimental design. Holding out the CEBPA master-regulator state as the target, the oracle reaches a cosine of 0.878 (residual norm 0.479) against a shuffled-target null with median 0.389 and 95th percentile 0.411 ($z \approx 37$ over 500 iterations), and a held-out-gene cosine of 0.856 ($z \approx 23.5$) (Fig. 4a). The recipe is led by CEBPE, the CEBPA paralog, with a knee at $k = 2$ (CEBPE + SPI1)—a biologically sensible minimal program. Where the target is not fully reached, the certificate again names an interpretable program: the CEBPA state requires raising a myeloid-differentiation module (MNDA, HP, VSIG4, NCF1) orthogonal to what the activation atoms in the screen supply.

Crucially, the Norman screen contains *measured* double perturbations, which let us test the additivity assumption that underlies every multi-gene recipe rather than merely assert it (Fig. 4b). Across 126 measured doubles, the cosine between the true double perturbation and the sum of its two singles has median 0.71: additivity is a good but *bounded* approximation, not an identity. We characterized the departure directly. The deficit grows with the combined perturbation magnitude (Spearman $+0.58$), consistent with a scale-free saturation ceiling ($M^\star = 13.9$ in units of the median single-effect norm, fit $R^2 = 0.57$); the intuitive hypothesis that collinear pairs should be most sub-additive was *refuted* (Spearman -0.16, not significant). The practical consequence, which we carry into every recipe claim, is that multi-gene knockdown recipes are an extrapolation to be tested, not an exact prediction, and their reliability degrades at large combined magnitude.

To probe generalization more systematically than a single held-out state, we applied the operator to an independent pair of cell types—the genome-scale essential-gene Perturb-seq screens of Replogle et al. (2022) in K562 and RPE1 cells (843 shared single-gene perturbations over 2,832 shared genes)—and asked, at three levels, what transfers across cell type (Supplementary Fig. S7). A single perturbation's effect *direction* transfers moderately: the matched cross-type cosine has median 0.35 against a shuffled-gene null of ≈ 0, rising to 0.64 along the common essential-stress axis. The reachability *verdict* transfers only as a graded quantity, not a binary flip—the residual norm is systematically worse under a cross-type basis than a within-type one ($0.61 \to 0.87$ for K562 targets, $0.37 \to 0.68$ for RPE1)—so a naive 100% verdict-agreement rate is an artifact of an over-expressive 843-atom cone and should be read as a residual, not a label. The minimal *recipe* does *not* transfer: the same-target Jaccard overlap across cell types is only 0.11, collapsing to the random-selection null (0.05) under a cross-type basis. The prescription is therefore bounded: the *direction* of a cell-state shift is portable across cell types, but the *specific minimal recipe* is cell-type-specific and must be re-derived per system. This is a cross-cell-type generalization probe on essential-gene screens, not a second $CD4^+$ T-cell polarization screen.

3.5. No prior method occupies the measured-effects, verdict-plus-certificate corner

We placed the oracle against a structured survey of 92 methods spanning 2011–2026 (Section 4), scored on two axes: how strongly each is grounded in measured effect vectors, and whether it emits a feasibility verdict with a certificate (Fig. 5a). The two capabilities are held by *disjoint* sets of prior methods. The 14 methods that operate directly on measured effects are all forward-only predictors or rankers; none returns a feasibility verdict or a data-grounded certificate. The 13–15 methods carrying any verdict- or certificate-like output are all model-theoretic—their guarantee is a property of a hand-built, fitted, or inferred model, and they span several families (structural and Boolean network control, control-from-data, an inferred single-cell GRN, a causally inspired predictor) rather than one. The intersection of "grounded in measured effects" and "emits a verdict plus certificate" is empty for all 91 prior methods; the reachability oracle is the sole occupant of that corner. The family-level capability matrix (Supplementary Fig. S1) shows the same structure per capability column: measured-effect grounding and feasibility output never co-occur in any prior family.

The claim is about the *combination* of two capabilities on a structured survey, not the unprecedentedness of either alone: network control has reasoned about reachability for over a decade, and many methods use measured data. The novelty is the pairing—a feasibility decision computed directly from measured effects, with a certificate—and the empty intersection is the evidence for it.

3.6. From verdict to experiment: a GO/REDIRECT/STOP triage

A feasibility verdict earns its keep by changing what experiment gets run. We applied the oracle to 102 real target nominations drawn from the T-cell transitions and translated each verdict into one of three actions (Fig. 5b): **GO**—the target is reachable, so hand the minimal ranked knockdown set to an arrayed CRISPRi screen; **REDIRECT**—the target is activation-required, so switch to a CRISPRa/agonist arm, with the certificate naming the genes to test first; and **STOP**—the target is provably outside the cone, so do not spend a knockdown arm on it. This matters because $45/102$ (44%) of the nominated targets are hard-to-drug and only 10 carry a clinical-grade drug, so misallocating screen arms is expensive. Across transitions, 36–49% (mean $\approx 42\%$) of each transition's addressable shift needs induction and is therefore unreachable by a knockdown-only screen—exactly the fraction a CRISPRi-only campaign would waste effort on without a feasibility check.

The triage surfaces concrete, and honestly caveated, examples. Nine targets are green-lit as directly reachable and druggable (including JAK2, ICOS, MAPK14, CD3D). Three are tractable but untried and flagged for a redirected arm (IL7R, which carries the highest genetic support; ZAP70; TET2). One, IRF1, is a top T_H2 knockdown node with strong immune-disease genetic support but no conventional small-molecule or antibody handle (degrader-only), a required-but-undruggable case where the verdict argues for a new modality rather than a conventional campaign. We emphasize that every one of these is a prioritization hypothesis for the bench, generated by geometry over one screen; none is a validated target, and the oracle's role is to triage where experimental effort is spent, not to replace the experiment.

4. Related work

Cell-state engineering poses an inverse-design question: given a starting state and a desired target state, which interventions move the cell there, and is the move even possible? This is the question behind directed differentiation, transdifferentiation, and T-cell engineering, and it is the question of pharmaceutical target selection. Getting it wrong is the dominant cost of drug development: only about one in ten clinical programmes reaches approval, and failure is paid for late (Minikel et al., 2024; Wong et al., 2019). Attrition concentrates at Phase II, and roughly half of Phase II/III failures are for lack of efficacy—the wrong target or mechanism rather than safety (Harrison, 2016). The two interventions with published evidence of improving those odds both point the same way: human genetic support raises the odds of approval about 2.6-fold (Minikel et al., 2024; Nelson et al., 2015), and *measured* rather than inferred target effects are the direct antidote to the 11–25% preclinical reproducibility documented by the Amgen and Bayer audits (Begley & Ellis, 2012; Prinz et al., 2011). A method for this problem should therefore (i) work from measured effects in the relevant cell type, (ii) answer achievability rather than merely rank candidates, and (iii) say so honestly when a goal is not achievable.

We surveyed 91 prior methods spanning 2011–2026 (92 including this work) across control-theoretic network control, gene-regulatory-network (GRN) inference with in-silico perturbation, deep perturbation-response prediction, single-cell foundation models, and the combinatorial-screen, optimal-transport, virtual-cell, and benchmark literature that borders them. Positioning them on two axes—*what question is answered* (forward prediction versus feasibility) and *what the answer is grounded in* (an inferred or hand-built model versus directly measured effects)—exposes a gap that is exact rather than rhetorical. The methods grounded in measured effects answer only forward questions, and the methods that reason about feasibility do so only inside a model.

Model-theoretic feasibility. The most mature theory for the intervention question is network control. Structural and target controllability cast steering an assumed linear system as a matching problem over network topology (Liu et al., 2011; Gao et al., 2014), and determining-node / control-kernel methods identify driver sets from structure (Mochizuki et al., 2013; Kim et al., 2013). Boolean attractor control computes node interventions that switch a logical network between attractors (Zañudo & Albert, 2015; Zañudo et al., 2017; Joo et al., 2018; Marazzi et al., 2022), a line that remains active (Cifuentes-Fontanals et al., 2025; Metzcar et al., 2025; Murrugarra et al., 2025; Trinh et al., 2025; Del Vecchio, 2026). A data-fitted strand fits a dynamical model to omics and then applies control theory (Ronquist et al., 2017; Rukhlenko et al., 2022; Wytock & Motter, 2024), and network-influence predictors rank transcription-factor sets for a conversion (Rackham et al., 2016; Cahan et al., 2014). These are the methods that carry a controllability or control-strategy guarantee—but the guarantee is a property of a hand-built, fitted, or inferred model, never of measured effects, and the control set is only as trustworthy as the network put in. The same dependency reappears whenever feasibility-style reasoning is imported onto a reconstructed single-cell GRN (Wang et al., 2023b) or run in a learned latent space (Han et al., 2025). The set of prior methods that emit any verdict- or certificate-like output is therefore not one family: it spans structural and Boolean network control, control-from-data, an inferred single-cell GRN, and one causally inspired perturbation predictor. What unites them is the assumed model, not the family.

Measured-grounded forward prediction. The single-cell community answers the same biological question by running an inferred network forward. CellOracle infers a GRN and simulates a transcription-factor knockout into predicted identity transitions (Kamimoto et al., 2023); related tools infer regulons, enhancer networks, or dynamic GRNs (Aibar et al., 2017; Xu et al., 2021; Bravo González-Blas et al., 2023; Wang et al., 2023a; Osorio et al., 2022), with BEELINE quantifying how method-dependent GRN inference is (Pratapa et al., 2020). Deep perturbation-response models are genuinely measured-grounded—they learn from real Perturb-seq data—but are resolutely forward: GEARS predicts outcomes of unseen multi-gene perturbations (Roohani et al., 2023), and compositional and disentanglement models predict responses to given perturbations (Lotfollahi et al., 2019; 2023; Hetzel et al., 2022; Piran et al., 2024). Single-cell foundation models (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024) and the 2025–2026 virtual-cell surge—perturbation transformers, atlas-scale models and challenges, and giga-scale perturbation atlases (Adduri et al., 2025; Roohani et al., 2025; Zhang et al., 2025; Klein et al., 2025)—extend the same forward map. Optimal-transport methods describe how cells *do* move rather than which interventions *would* move them (Schiebinger et al., 2019; Lange et al., 2022), and experimental-design methods choose the next screen (Huang et al., 2024). Every one maps perturbation to predicted response: a predictor always returns *some* answer and structurally cannot certify that a target is unreachable.

Why a feasibility verdict, not another predictor. A cluster of 2025–2026 benchmarks reports that current perturbation-prediction models, foundation models included, often fail to beat simple mean or linear baselines on genuinely unseen perturbations (Ahlmann-Eltze et al., 2025; Wong et al., 2025; Csendes et al., 2025; Kedzierska et al., 2025; Wu et al., 2025; Viñas et al., 2025; Kernfeld et al., 2025; Wei et al., 2025; Luecken et al., 2025; Atti & Subramaniam, 2025; Mao et al., 2026; Vollenweider & Bühlmann, 2026). This is the field's own critique, and it sharpens the case for a different output: a feasibility verdict computed directly from measured effects does not extrapolate to unseen perturbations at all, and it comes with a checkable certificate and a shuffled-target null. The claim is deliberately weaker, and therefore more defensible, than predicting an unseen response.

Our position. Across all 91 prior methods (ours excluded), 14 operate directly on measured effects—and all 14 only predict or rank; *none* returns a feasibility verdict, emits a data-grounded certificate, or proves infeasibility, and the 13–15 partial verdict/certificate marks are all model-theoretic. The two capabilities this work combines—measured grounding and a feasibility verdict with a certificate—are held by *disjoint* sets of prior methods, and their intersection is empty. We compose the measured genome-scale CRISPRi effect vectors from a Perturb-seq screen in primary human CD4$^+$ T cells (Zhu et al., 2025) directly, with no inferred network. Because knockdown is loss-of-function and combinations are non-negative, reachability becomes membership in a convex cone, solved by non-negative least squares (Lawson & Hanson, 1974): "reachable" returns the mixing weights (a minimal ranked recipe), and "outside the cone" is a theorem whose separating hyperplane—a Farkas/KKT certificate (Farkas, 1902)—names the genes that would have to be *activated* instead, a falsifiable CRISPRa hypothesis rather than a null result. We validate by held-out-gene generalisation against a shuffled-target null (Th2→Th1 held-out cosine 0.448, null $z \approx 24$; KKT residual $\approx 10^{-11}$), and the operator transfers unchanged to the Norman K562 combinatorial CRISPR screen (Norman et al., 2019) (held-out CEBPA cosine 0.878), whose measured double-perturbations show that the additivity assumption is a bounded approximation (median cosine 0.71, sum-of-singles versus measured) rather than exact. We claim novelty for this measured-effect, certificate-carrying reachability *decision*, not for the T-cell or CEBPA regulator lists, which the source screens (Zhu et al., 2025; Norman et al., 2019) already report and which we use only as positive controls; activation hypotheses require a separate CRISPRa arm (Schmidt et al., 2022), and the modest absolute cosine is the ceiling of loss-of-function alone.

5. Discussion

We have described a feasibility oracle that treats a genome-scale CRISPRi screen as a dictionary of measured effect vectors and decides, by convex geometry, whether a target cell-state shift can be reached by a non-negative combination of knockdowns. The decision is constructive on both sides: a reachable target returns a minimal ranked recipe, and an unreachable one returns a Farkas/KKT certificate that names the genes requiring activation. The value of the framework is not a new list of T-cell regulators—those are already in the source screens and we use them only as positive controls—but a new *kind of output*: a falsifiable feasibility verdict computed directly from measured effects, which no method in a 92-entry survey provides.

5.1. Interpretation

The central biological finding is that a real polarization shift ($T_H2 \rightarrow T_H1$) is only *partly* reachable by knockdown (held-out cosine 0.448; 39% of the shift in the loss-of-function cone), and the same pattern recurs across the 12-cell atlas, where knockdown never accounts for the majority of a transition (mean loss-of-function fraction 0.34). A knockdown-only campaign, however well designed, therefore addresses a minority of most cell-state transitions; the remaining majority is a mix of required activation and an irreducible remainder. This is a property of the modality, not a shortfall of fit—the headline cosine is 71% of the ceiling a knockdown-only cone can reach in principle (atlas mean 61%; equivalently, as that ceiling coincides with the in-sample cosine, the held-out fit recovers 71% of the in-sample one, not an independent bound)—which is why a single reachability score would be misleading and we report a signed decomposition instead.

The infeasibility certificate is what makes this limitation actionable: it names, reproducibly, the genes carrying the unmet upward demand, so a "no" redirects an experiment rather than ending it. Its reading has a specific caveat, though. The highest-demand genes are state-markers and *negative* regulators, not the canonical master transcription factors (which are present but rank deep), so the certificate is best read as a reproducible statement of *where* activation is required, not a validated list of activation targets—and for a negative regulator the useful intervention may be repression, the opposite of naive activation. Finally, the operator transfers unchanged to other cell types, but only in part: the *direction* of a shift is portable while the *specific minimal recipe* is not, so the feasibility geometry generalizes even where its quantitative prescription must be re-derived per system.

5.2. Limitations

Three limitations bound the conclusion; a fuller, item-by-item reinforcement plan accompanies the manuscript as a standalone document.

The most consequential is that every multi-gene recipe assumes co-perturbation effects add. On the 126 measured doubles available (Norman K562) the measured double aligns with the sum of its singles at median cosine 0.71—bounded, not exact, and degrading with combined perturbation magnitude (Spearman $+0.58$; scale-free saturation ceiling $M^\star = 13.9$) rather than with collinearity (Spearman -0.16, not significant). Recipes beyond a few knockdowns, or of individually large effect, are therefore extrapolations whose error is real, and the $CD4^+$ T-cell dictionary contains no in-domain doubles against which to check them.

The verdict's second exposure is not random measurement error but coordinated bias. Resampling each effect within its standard error and re-solving leaves the polarization verdicts intact—undirected noise, if anything, inflates reachability by adding anisotropy the null already corrects for—whereas a coordinated attenuation of only $\approx 8\%$ of the target-aligned signal (≈ 0.03 standard-error units) suffices to flip the T_H1 verdict. This is exactly the failure mode a randomized interventional design excludes, which is what makes the design-based identification load-bearing rather than incidental; the same analysis flags the imported CD4 aging axes as fragile at the differentiated state, where at Stim48hr they fall below the anisotropy null and should not be read as reachable. External validity is otherwise bounded by provenance: the primary results come from one primary-cell system with four donors, collapsed across donors, so a true leave-one-donor-out test is impossible here and cross-genotype and cross-tissue generality rests on the K562 transfer rather than the primary system.

The statistical scaffolding, finally, answers narrower questions than the word "reachability" suggests. The shuffled-target null shows only that a target beats a structure-matched random direction, not that its recipe is biologically correct; held-out-gene validation generalizes across expression coordinates, not across cells or conditions; and because the in-sample cosine overstates the held-out value (mean gap 0.225 across the atlas), every headline number here is the held-out one. The certificate ranking is reproducible under a gene-axis split (cross-half Spearman $\rho \approx 0.65$, half-versus-full ≈ 0.84), but its biological reading—like the transcription-factor-versus-marker split in the positive control—is retrospective and post-hoc, and is offered as annotation rather than validation.

Beneath all three is a proxy assumption: the target is a transcriptomic signature, and matching it in expression space is not the same as achieving the corresponding cellular phenotype—functional confirmation (cytokine output, chromatin state, *in vivo* function) remains a separate wet-lab endpoint.

5.3. Outlook

The most valuable next experiments follow directly from the limitations. Measuring a modest panel of double knockdowns in the primary $CD4^+$ system would replace the borrowed K562 additivity estimate with an in-domain one and calibrate the recipe-size ceiling. Running one CRISPRa arm against a certificate's named genes would provide the first direct test of an infeasibility prediction—the framework's most distinctive and least-validated claim. And extending the dictionary to additional cell types and donors would convert the transfer result from an existence proof into a generalization curve. In each case the framework's output is what makes the experiment efficient: it says not only what to test but where testing is unnecessary. We therefore position the oracle as an experiment-triage instrument—a way to decide where a screen should spend its arms and where it should stop—rather than a target-validation engine, and we release it in that spirit.

A causal-inference agenda. Read as a causal-inference method, the oracle is a design-based counterfactual query: the effect vectors are average treatment effects from a randomized CRISPRi experiment, and a reachability verdict asks whether a non-negative compound intervention exists that points at the target state. That framing makes the identifying assumptions explicit and turns each into a concrete next experiment. Four are worth naming. First, *interference* (SUTVA): pooled Perturb-seq lets a perturbed cell's secreted cytokines alter a neighbour's state, which would bias the very polarization effects the target axis is built from; an arrayed-versus-pooled or MOI-titration screen would bound this spillover directly and is the primary external-validity test. Second, *mediation*: decomposing a recipe's effect through master-regulator intermediates would distinguish a direct lever from one acting only through a downstream factor, sharpening the minimal recipe into a mechanistic one. Third, *dynamics*: the current verdict is static, whereas differentiation is a trajectory, so a time-resolved dictionary with g-methods (accounting for time-varying confounding) would convert a reachability verdict into a reachable *schedule*. Fourth, *transportability*: the aging axis in particular is a direction imported from a different population, so a selection-diagram analysis would state which effects are licensed to transport and re-express the aging verdict as a transported rather than assumed-invariant quantity. The computable half of this agenda—sensitivity radii, negative-control outcomes, signed construct validity, weak-instrument-robust recipe intervals, and the instrumental-variables treatment of guide non-compliance—is already implemented as a causal-validation dossier accompanying the code; the four experiments above are its wet-lab continuation.

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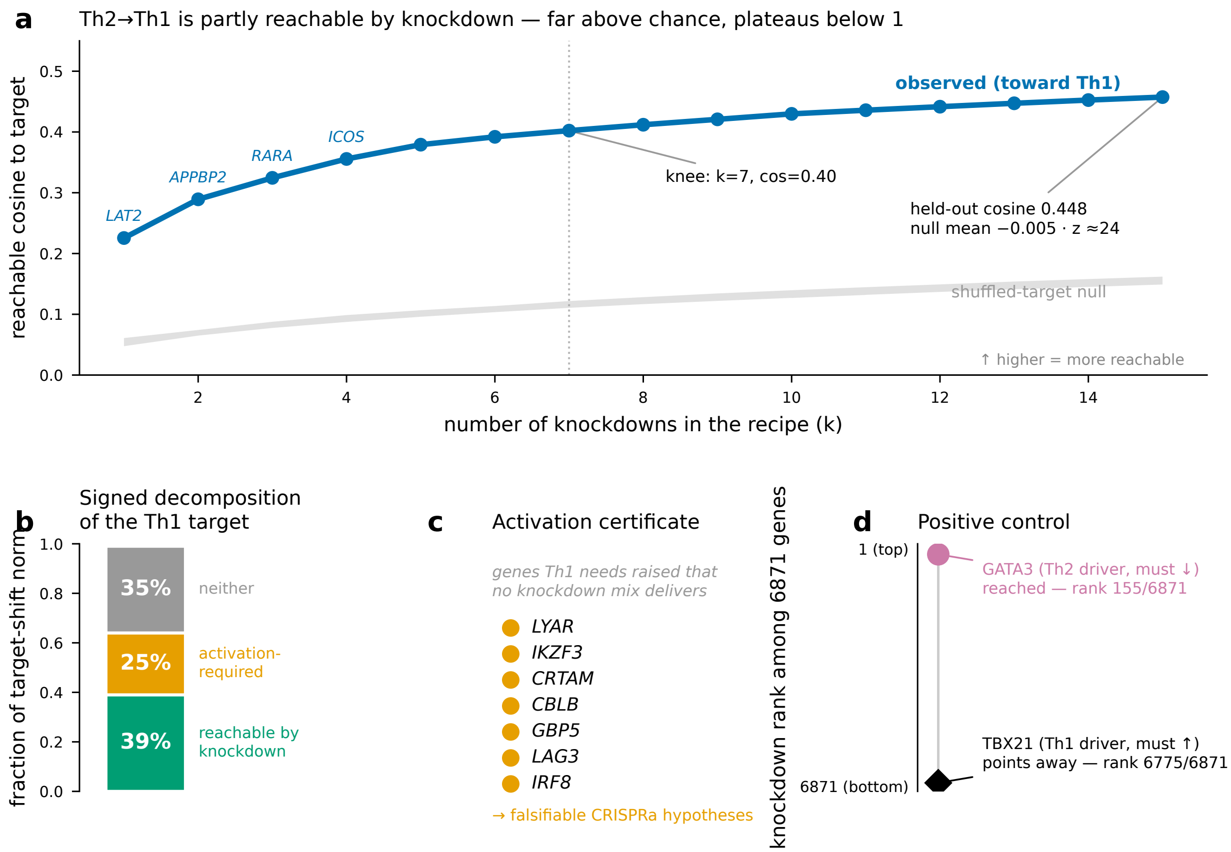

Figure 2. **$T_H2 \to T_H1$ is partly reachable by knockdown.** **(a)** Greedy reachability spectrum: reachable cosine versus recipe size k, far above the shuffled-target null at every k, with a knee at $k = 7$ (cosine 0.40) and a held-out cosine of 0.448 (null mean -0.005, $z \approx 24$). **(b)** Signed decomposition of the target: 39% reachable by knockdown, 25% activation-required, 35% neither. **(c)** Activation certificate: seven genes whose expression the T_H1 target needs raised but no knockdown mix supplies—falsifiable hypotheses in expression space, with the realizing intervention differing by gene role (activation for effectors, de-repression for negative regulators such as IKZF3/CBLB). **(d)** Positive control: the T_H2 driver GATA3 is reached (rank $155/6871$) while the T_H1 driver TBX21 is correctly anti-aligned (rank $6775/6871$).

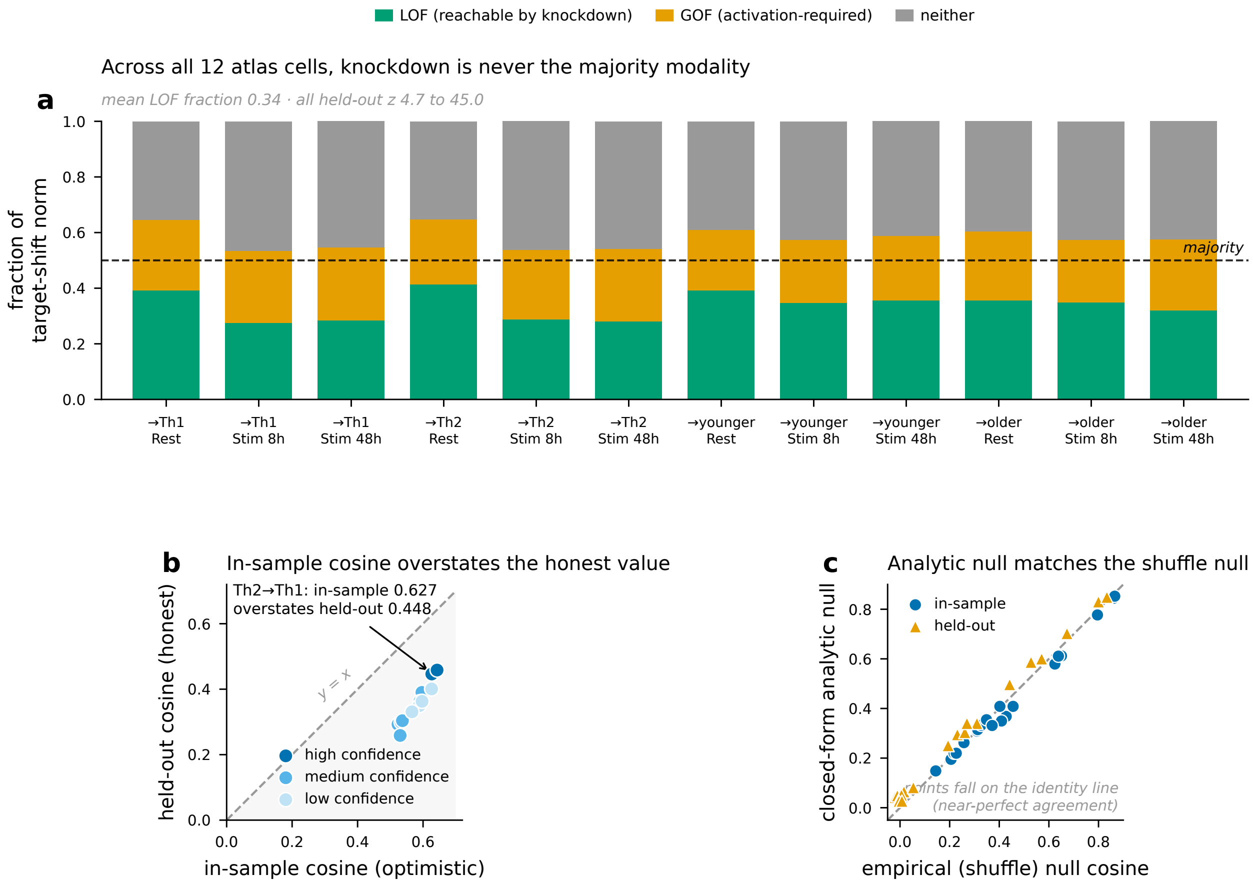

Figure 3. **Across a 12-cell atlas, knockdown is never the majority modality. (a)** Signed LOF/GOF/neither decomposition for all 12 targets (four transitions $\times$ three activation states); the loss-of-function (green) bar never crosses the 50% majority line, mean LOF fraction 0.34, all held-out z between 4.7 and 45.0. **(b)** In-sample reachable cosine overstates the held-out cosine on every transition (points below $y = x$), mean gap 0.225; marker shade encodes the confidence tier. **(c)** The closed-form analytic null matches the empirical shuffle null (near-identity), justifying its use atlas-wide.

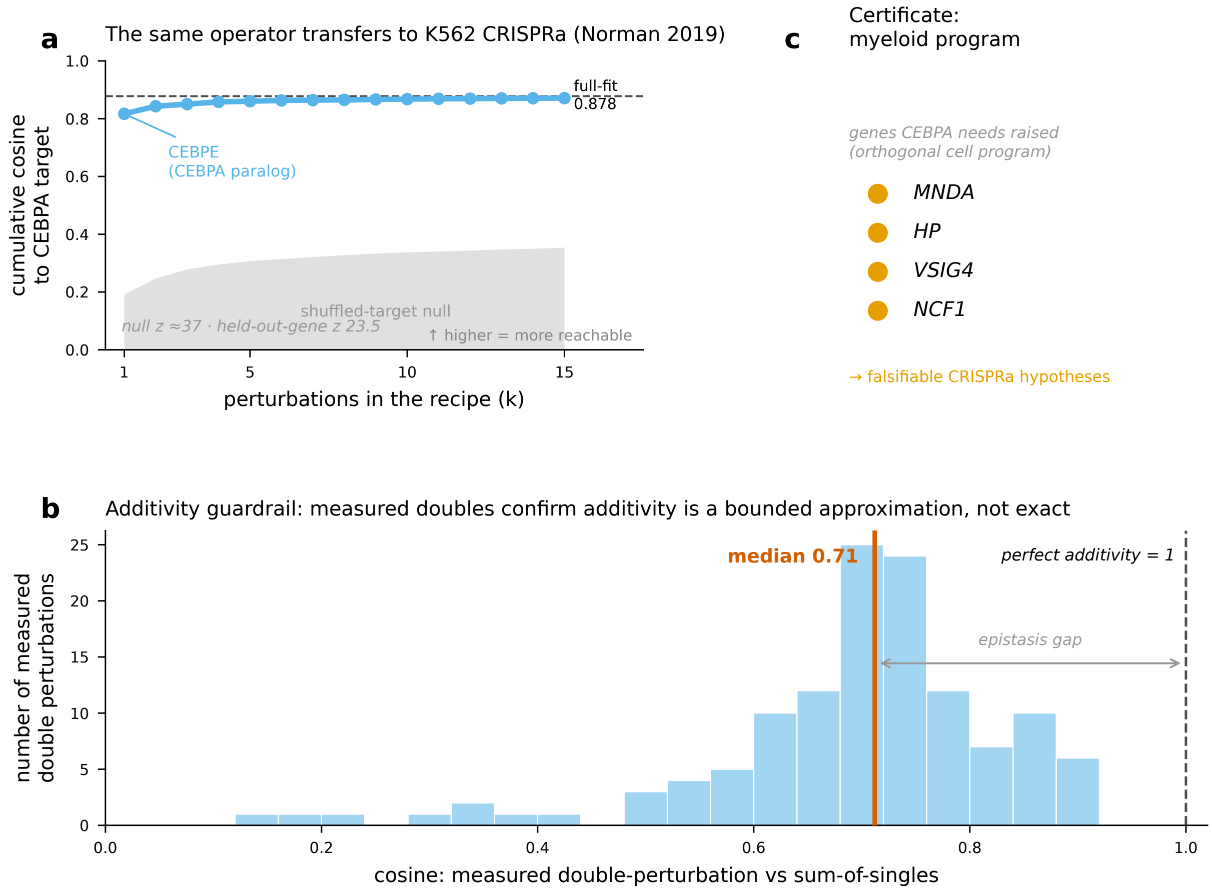

Figure 4. **Transfer to Norman 2019 K562 CRISPRa and the additivity guardrail. (a)** The unchanged operator reaches the held-out CEBPA state at cosine 0.878 (full-fit), far above the shuffled-target null ($z \approx 37$; held-out-gene $z \approx 23.5$), with the recipe led by the CEBPA paralog CEBPE. **(c)** The infeasible remainder's certificate names a myeloid program (MNDA, HP, VSIG4, NCF1). **(b)** Across 126 measured double perturbations, the cosine of the measured double to the sum of its singles has median 0.71: additivity is a bounded approximation, and the "epistasis gap" to perfect additivity (= 1) is real.

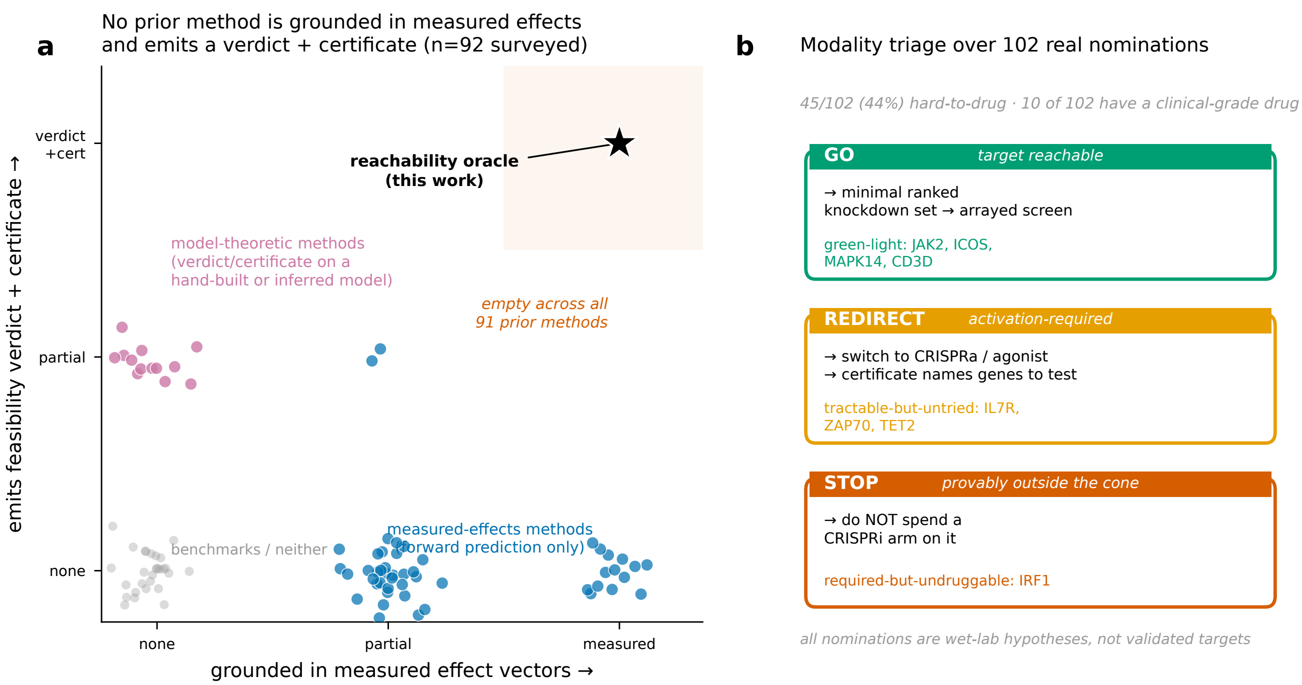

Figure 5. **Landscape positioning and modality triage.** **(a)** Each of 92 surveyed methods (2011–2026) placed by measured-effect grounding (x) and feasibility-verdict-plus-certificate output (y). Measured-grounded methods (blue) are forward-only; verdict/certificate methods (pink) are model-theoretic; the top-right corner is empty across all 91 prior methods and occupied only by this work. **(b)** The verdict translates into a GO/REDIRECT/STOP triage over 102 real target nominations ($45/102 = 44\%$ hard-to-drug, 10 with a clinical-grade drug); all nominations are falsifiable wet-lab hypotheses, not validated targets.

A. Supplementary figures

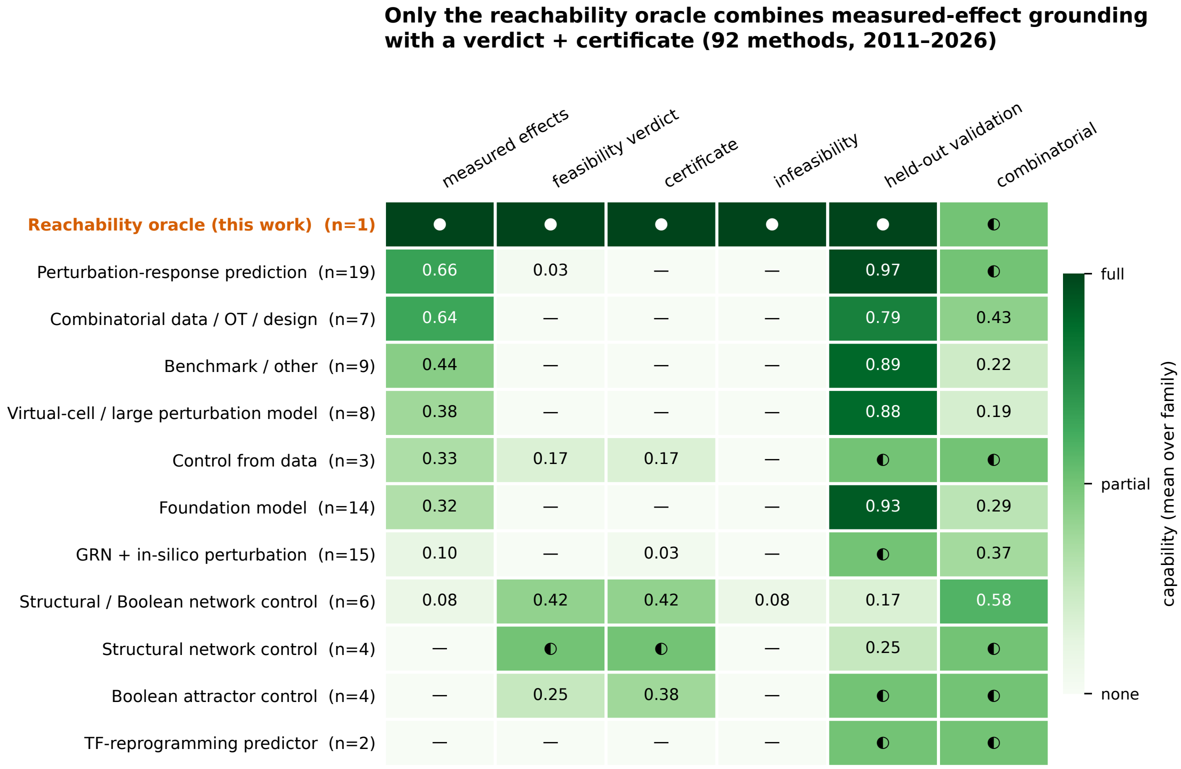

Figure S1. **Family-level method × capability matrix.** Mean capability over each method family for six capabilities—measured-effect grounding, feasibility verdict, certificate, infeasibility proof, held-out validation, and combinatorial support—across the 92 surveyed methods (2011–2026). The reachability oracle (top row) is the only entry that combines full measured-effect grounding with a feasibility verdict, certificate, and infeasibility proof; every prior family that scores on the verdict/certificate columns scores near zero on measured-effect grounding, and vice versa.

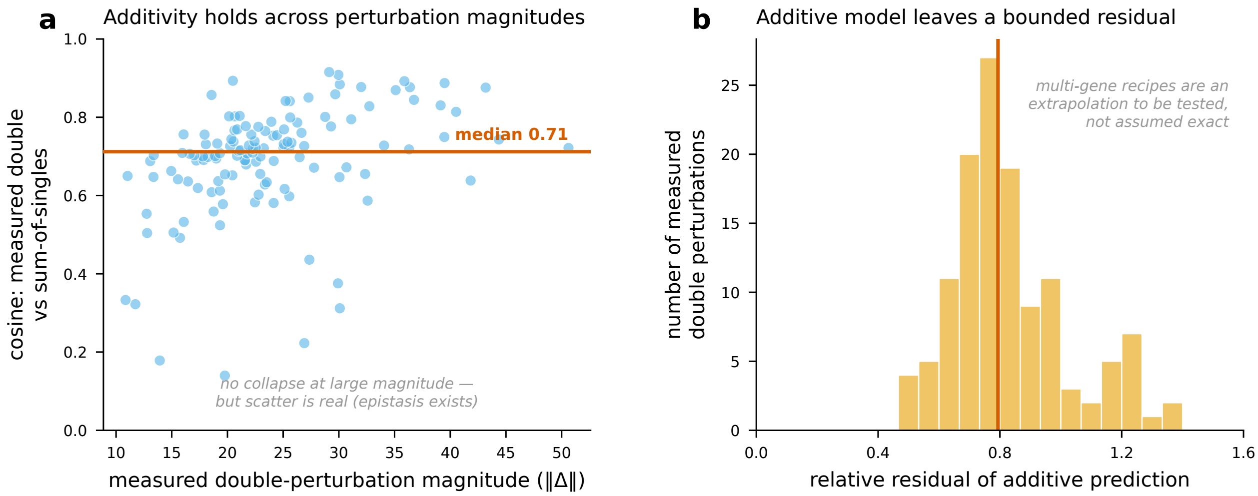

Figure S2. **Additivity is a bounded approximation across perturbation magnitudes. (a)** For the 126 measured double perturbations in the Norman K562 screen, the cosine between the measured double and the sum of its two singles (median 0.71) does not collapse at large combined perturbation magnitude, but the scatter is real—epistasis exists. **(b)** The relative residual of the additive prediction is bounded and centered below one, so multi-gene recipes are an extrapolation to be tested at the bench rather than an exact prediction.

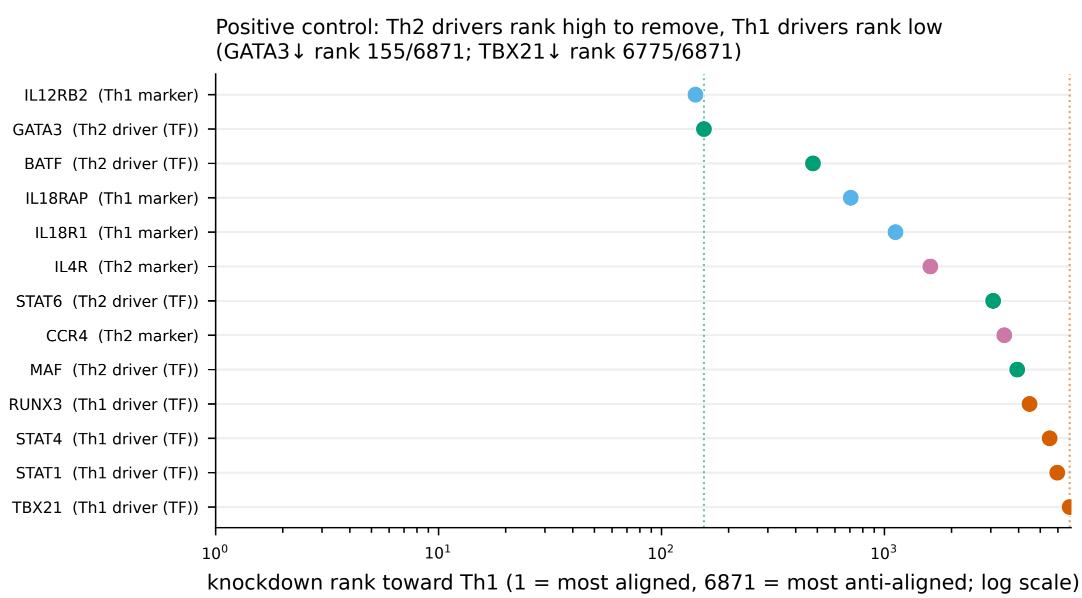

Figure S3. **Positive-control ranking of T-helper regulators.** Knockdown rank toward the T_H1 target (log scale; 1 = most aligned with the loss-of-function direction, 6871 = most anti-aligned) for canonical T_H1 and T_H2 regulators. T_H2 drivers, which must be removed, rank high (GATA3 at 155/6871); T_H1 drivers, which must be raised, rank low (TBX21 at 6775/6871). The transcription-factor-versus-marker split is an exploratory, post-hoc annotation, not a pre-registered test.

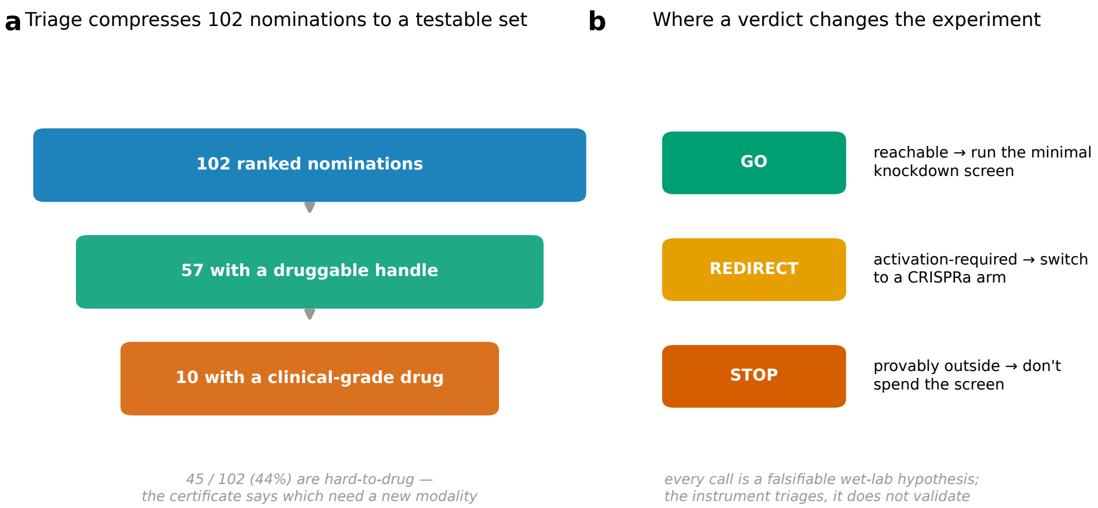

Figure S4. **Triage funnel and the three actions a verdict enables.** **(a)** The oracle compresses 102 ranked nominations to a testable set: 57 carry a druggable handle and 10 have a clinical-grade drug, while $45/102$ (44%) are hard-to-drug. **(b)** Each verdict maps to one action—GO (reachable, run the minimal knockdown screen), REDIRECT (activation-required, switch to a CRISPRa arm), or STOP (provably outside, do not spend the arm). Every call is a falsifiable wet-lab hypothesis; the instrument triages, it does not validate.

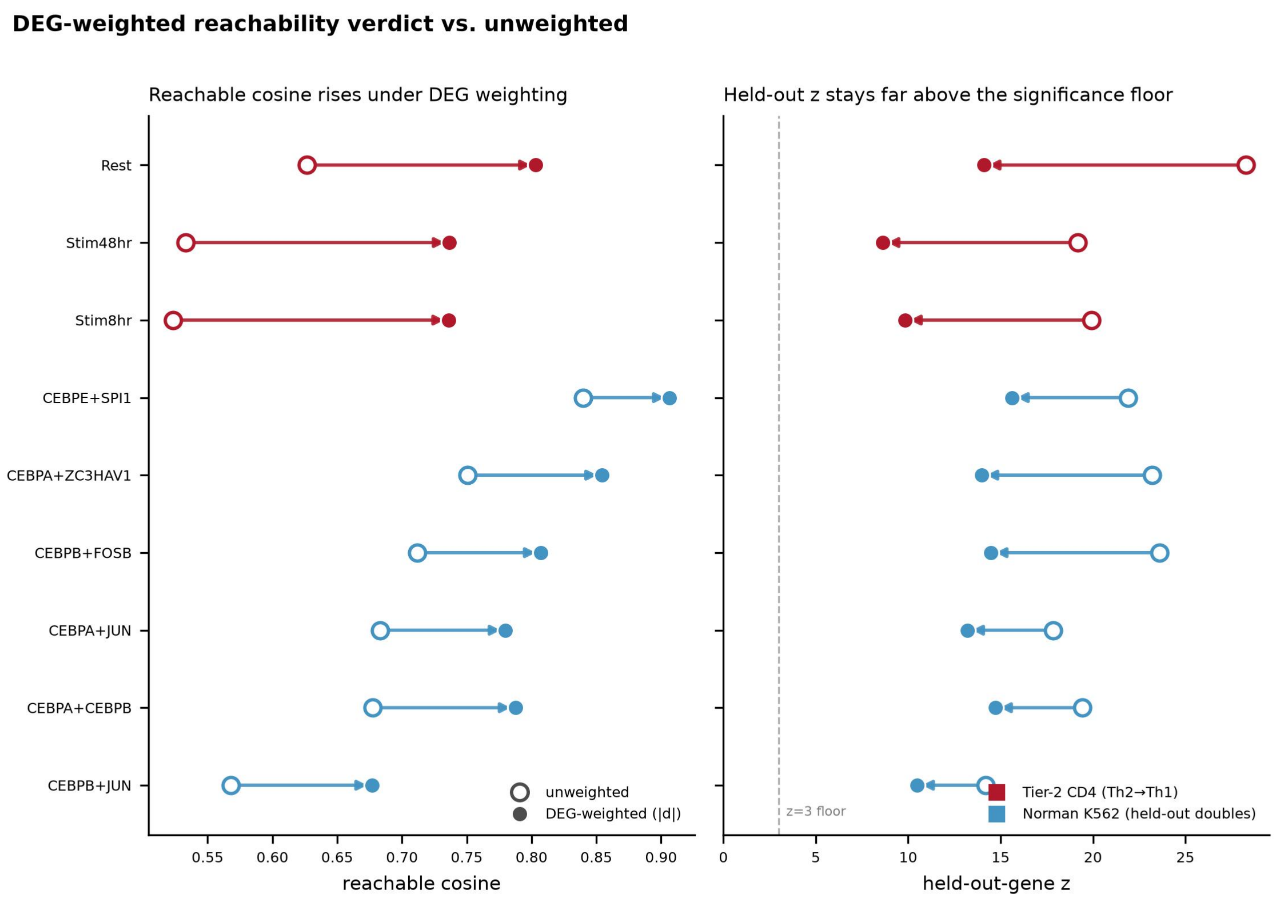

Figure S5. **The verdict strengthens under DEG-magnitude weighting.** For each target, the reachable cosine **(left)** and the held-out-gene z **(right)** under the default uniform metric (open markers) and under a DEG-magnitude weighting $w_g = |d_g|$ that concentrates the metric on the genes the target actually moves (filled markers). Reweighting *raises* the reachable cosine for every target—the Tier-2 $\mathrm{T_H}2 \to \mathrm{T_H}1$ shift rises from 0.627 to 0.803 at rest—and the held-out z stays far above the $z = 3$ significance floor, so the signal is not diluted by unchanged genes. This addresses the signal-dilution critique of Mejia et al. (2025); the unweighted metric remains the default and reproduces the published numbers exactly.

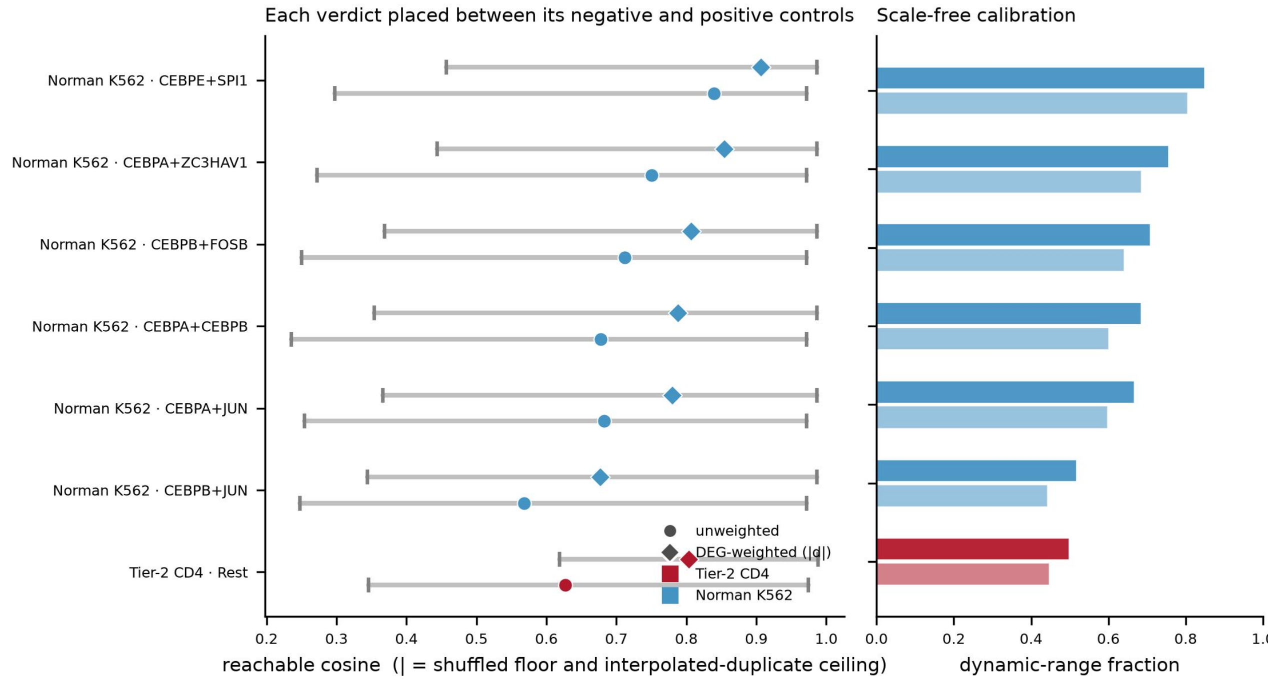

Figure S6. **Each verdict is calibrated between its shuffled floor and an interpolated-duplicate ceiling. (left)** For each target, the observed reachable cosine (unweighted, circle; DEG-weighted, diamond) placed between a shuffled-gene floor and an interpolated-duplicate ceiling (whiskers). Both floor and ceiling rise under weighting, so the observed value's placement between them is stable. **(right)** The scale-free dynamic-range fraction—the observed cosine expressed as a position between floor and ceiling—barely moves under weighting (Tier-2 rest $0.447 \to 0.499$), confirming that the strengthening in Fig. S5 is a genuine, calibrated signal rather than a scaling artifact. The dynamic-range calibration is computed on the rest condition (Monte-Carlo cost).

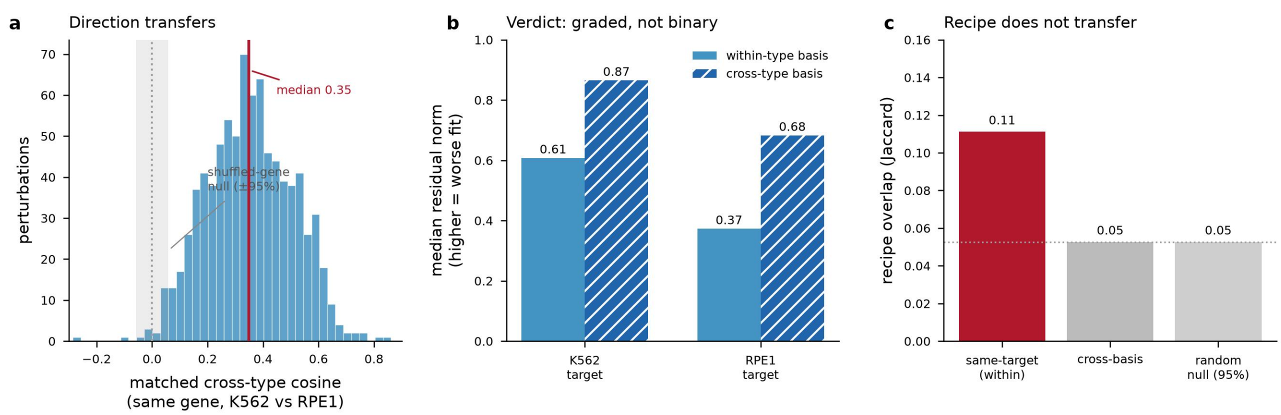

Figure S7. **Cross-cell-type transfer: direction transfers, the minimal recipe does not.** The unchanged operator applied to the Replogle et al. (2022) K562 and RPE1 essential-gene screens (843 shared perturbations, 2,832 shared genes). **(a)** A single perturbation's effect *direction* transfers moderately (matched cross-type cosine median 0.35; grey band, shuffled-gene null $\pm 95\%$). **(b)** The reachability *verdict* transfers only as a graded residual: the median residual norm is systematically worse under a cross-type basis (hatched) than a within-type basis, for both K562 and RPE1 targets. **(c)** The minimal *recipe* does not transfer—same-target Jaccard overlap (0.11) collapses toward the random-selection null (0.05) under a cross-type basis. Direction is portable across cell types; the specific recipe is not.

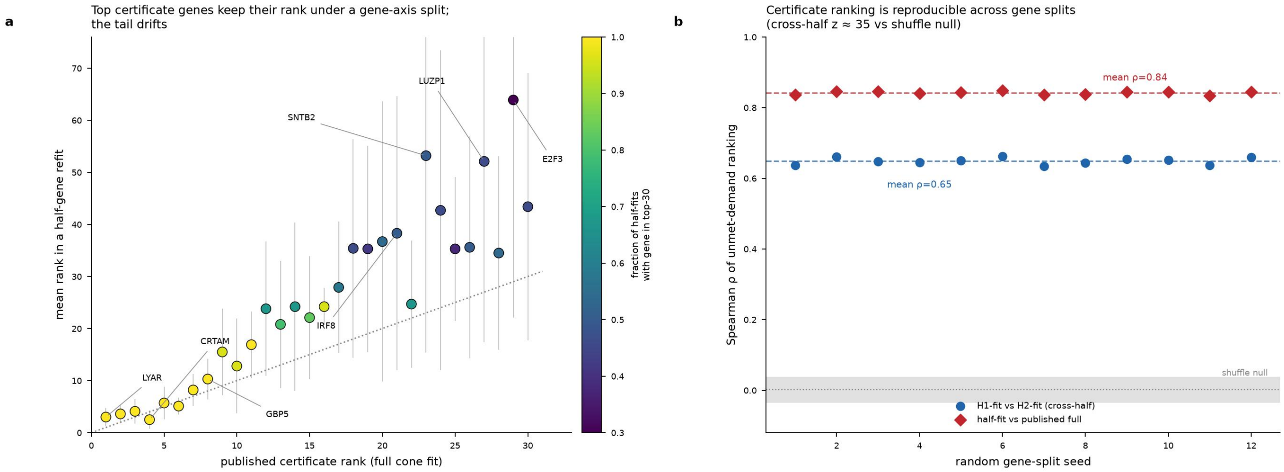

Figure S8. **The activation certificate is reproducible under a gene-axis split. (a)** Mean rank of each top certificate gene when the certificate is refit on a random half of the gene axis (vertical bars, spread over 12 splits), against its published rank on the full cone fit (colour, fraction of half-fits keeping the gene in the top 30). The highest-demand genes (LYAR, CRTAM, GBP5, IRF8) hug the identity line; the tail (E2F3, LUZP1, SNTB2) drifts. **(b)** Rank agreement across independent halves (cross-half Spearman $\rho \approx 0.65$) and between a half-fit and the full fit ($\rho \approx 0.84$) is stable across 12 random split seeds and far above the shuffle null (grey band; cross-half $z \approx 35$).

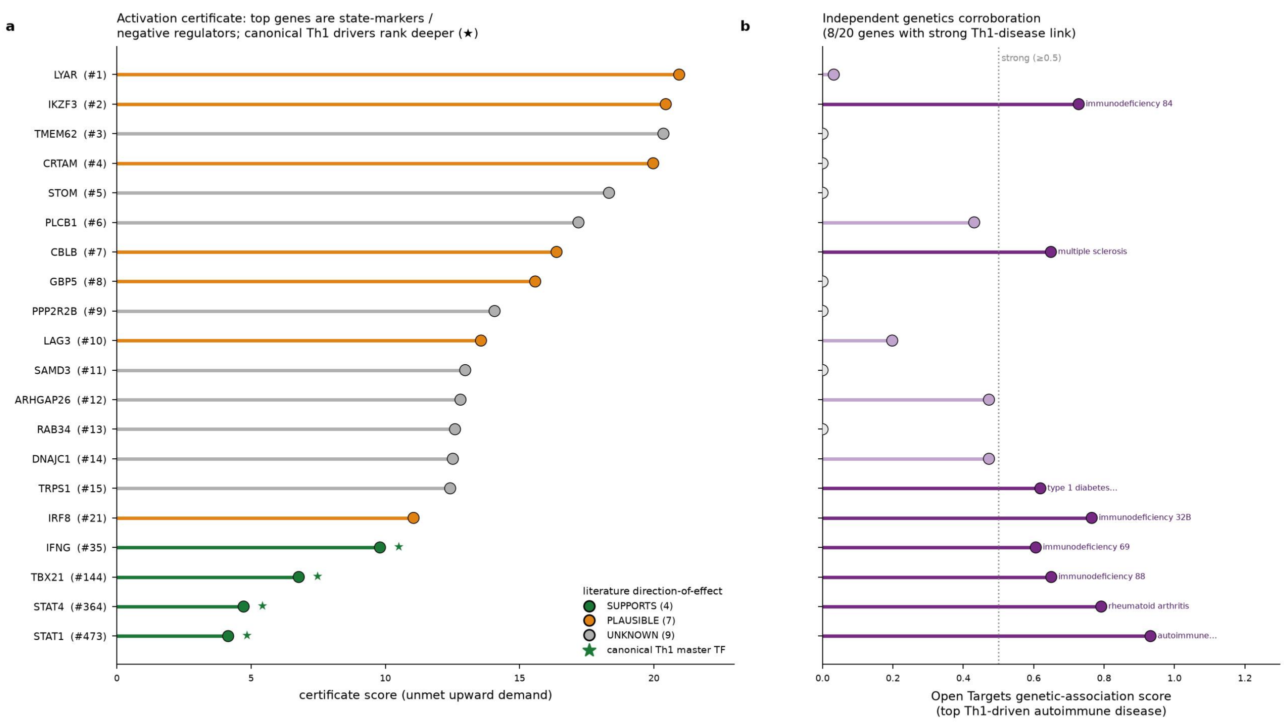

Figure S9. **Retrospective cross-reference of the top certificate genes tempers their biological reading. (a)** The top 20 certificate genes ranked by unmet upward demand, coloured by literature direction-of-effect (SUPPORTS 4, PLAUSIBLE 7, UNKNOWN 9). The highest-demand genes are state-markers (LYAR, CRTAM, GBP5) and *negative* regulators (IKZF3/Aiolos, CBLB, LAG3), while the canonical T_H1 master transcription factors (green stars) are present in the certificate but rank deep (IFNG #35, TBX21 #144, STAT4 #364, STAT1 #473). **(b)** Independent genetic corroboration: $8/20$ genes carry a strong (≥ 0.5) Open Targets genetic-association score to a T_H1-driven autoimmune disease. The certificate is best read as a reproducible ranking of unmet upward demand, not a validated list of activation targets; for the negative regulators, a useful intervention may be opposite in sign to naive activation. This cross-reference is exploratory and post-hoc.